



Exploring microalgae-based strategies for the management of white spot syndrome virus: a comprehensive review

Amrutha Vellore Mohankumar¹ · Vernita Priya¹ · Sanjana Anbalagan¹ · Vidya Radhakrishnan² · Jayakrishna Kandasamy³ · Tohru Mekata⁴ · Sudhakaran Raja¹

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Abstract

Shrimp culture industries faced huge economic losses due to white spot disease (WSD) caused by the white spot syndrome virus (WSSV). Microalgae are unicellular and can be found in both freshwater and marine habitats. In recent days, microalgae such as *Chlamydomonas reinhardtii*, *Dunaliella salina*, *Synechocystis species*, *Synechococcus elongatus*, *Synechococcus species*, and *Anabaena* species have been explored for treating WSSV using various protocols including transgenic algae, recombinant protein, and dsRNA. The development of dsRNA-expressing lines in *C. reinhardtii* has improved the efficiency of the new vectors p2xTRBL and pSS116, enabling the selection of transgenic cells without needing markers. Producing VP28 dsRNA to fight the white spot syndrome virus (WSSV) in shrimp has proven particularly advantageous. Due to the development of recombinant plasmids that express VP28 genes in microalgae, oral vaccinations against WSSV have proven highly effective and demonstrated great survival rates in shrimp. Moreover, *Dunaliella salina* transgenic strains have been modified to generate VP28 genes to shield crayfish from WSSV. Because of their antiviral qualities against WSSV, microalgae such as *Synechococcus* sp. and *Anabaena* sp. have also been used and may be useful in managing other aquatic viral infections. Through oral vaccination using sustainable and environmentally friendly practices, these microalgae serve as a disease-control strategy in aquaculture, improving the financial standing and income of aquaculture farmers. All microalgae have antiviral properties and act against the white spot syndrome virus (WSSV).

Keywords WSSV · Microalgae · Transgenic algae · Recombinant protein · dsRNA

Abbreviations

WSD White spot disease
WSSV White spot syndrome virus
dsRNA Double standard ribonucleic acid

FAO Food and Agriculture Organization
FDA Food and Drug Administration
PL Post-larvae
TAP Tris-acetate-phosphate
PCR Polymerase chain reaction
RT-PCR Real-time polymerase chain reaction
ELISA Enzyme-linked immunosorbent assay
LSZ Lysosome
ALP Anti-lipopolysaccharides
SOD Superoxide dismutase
ProPO Prophenoloxidase

Amrutha Vellore Mohankumar and Vernita Priya these two authors contributed equally to this work.

✉ Sudhakaran Raja
sudhakaran.r@vit.ac.in

- ¹ Aquaculture Biotechnology Laboratory, School of Biosciences and Technology (SBST), Vellore Institute of Technology, Vellore, Tamil Nadu 632014, India
- ² VIT School of Agricultural Innovations and Advanced Learning (VAIAL), Vellore, Tamil Nadu 632014, India
- ³ School of Mechanical Engineering, Vellore Institute of Technology, Vellore, Tamil Nadu 632014, India
- ⁴ Research Institute of Frontier Science and Technology, Okayama University of Science and Technology, Imabari-shi, Japan

Introduction

According to the UN Food and Agriculture Organization (FAO), global aquaculture production of crustaceans was 223.2 million tons in 2022, accounting for 4.4% of the global aquaculture production. Shrimp aquaculture is the

fastest-growing animal food-producing sector in the world due to its high demand in developed countries (Karthikeyan et al. 2022) (Fig. 1).

Over 9.4 million tons of shrimp are produced annually worldwide mainly for the highly profitable shrimp aquaculture industry. However, in recent years, output has struggled significantly, primarily due to the persistent prevalence of illnesses, which has negatively impacted the growth of this sector. The disease response has resulted in a slow recovery of the shrimp industries in the affected countries (El-Saadon et al. 2022).

White spot disease (WSD), caused by the white spot syndrome virus (WSSV) in penaeid shrimp, has a major negative economic impact on the shrimp culture sectors (Priya et al., 2024). WSD in crustaceans particularly affects cultivable penaeid species including *Penaeus monodon*, *P. vannamei*, *Marsupenaeus japonicus*, and *Fenneropenaeus indicus*. WSSV is a global hazard to industries and shrimp producers (Sánchez-Paz et al. 2010).

WSSV is a rod-shaped, encapsulated dsDNA virus with a tail-like appendage about 275 nm long and 120 nm wide. The genome of this virus roughly ranges from 280 to 315 kbp in size and has 184 open reading frames (Kawato et al. 2023). The first report of WSSV infection was from Taiwan in the early 1990s, followed by other countries. The aquaculture of shrimp suffered a severe economic loss in the latter. WSSV causes the infection between 2 to 10 days with a 100% mortality rate. The fact that there is presently no effective prevention and control for WSSV infection is likely because shrimp lacks true adaptive immunity. Since there is no reported treatment with medication for WSSV, control measures are urgently required (Oakey et al. 2019).

Microalgae are unicellular microorganisms ranging in size from very small (a few microns) to big and can be found in freshwater and marine habitats (a few hundred microns) (Gul et al. 2024). With the help of solar energy, carbon dioxide, and nutrients such as nitrogen, sulfur, and phosphorus, they can create biomass quickly. Compared to

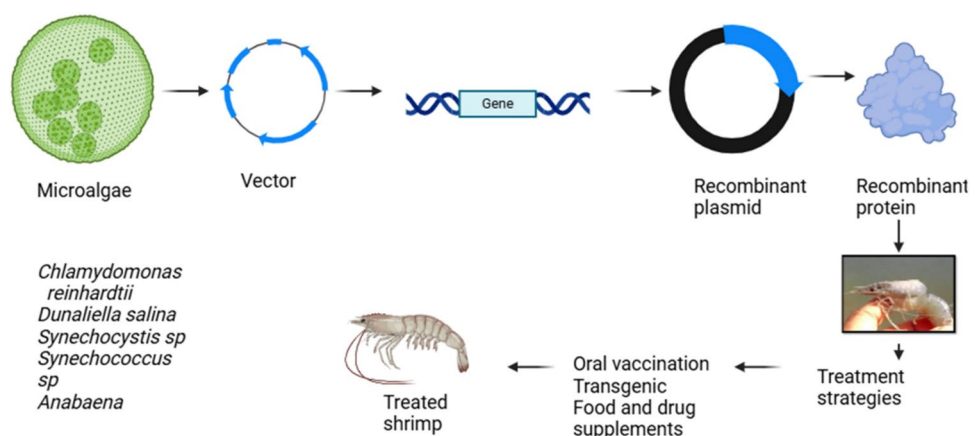
other bioresources, simple maintenance and culture in artificial settings provide a profitable platform to manufacture and extract bioactive molecules from microalgae. Microalgae have recently drawn more scientific attention because of their adaptable development, useful metabolic characteristics, and capacity to produce biopharmaceuticals. The unique qualities of microalgae have led biotechnologists to develop advanced genetic and molecular ways to employ them as green bioreactors for biopharmaceuticals, a classification that the Food and Drug Administration (FDA) calls “safe”. *Anabaena* species, *Dunaliella salina*, *Synechocystis* sp., *Synechococcus*, and *Chlamydomonas reinhardtii* are important components of animal diets, particularly those including shrimp and fish larvae (Xavier et al. 2023).

Chlamydomonas reinhardtii, a unicellular microalga, has substantial potential as a light-driven cell factory for producing recombinant proteins like subunit vaccines. Due to its benefits as a secure and reliable platform, green microalgae *Chlamydomonas reinhardtii* have recently been employed extensively to create edible vaccines and pharmaceutical proteins for aquaculture (Charoonnart et al. 2022).

Possessing basic unicellular structure and ease of cultivation, *Dunaliella salina*, a eukaryotic green alga, has been used as a new bioreactor. Under natural conditions, it is capable of large-scale growth and can efficiently accumulate protein products with high yields. The isolated, large, cup-shaped chloroplast found in *D. salina* cells can be exploited as a more efficient alternative to nuclear expression for example, “position effect” or “gene escaping” and it is possible to stop “gene silencing”. *Dunaliella salina* is the best selection for a bioreactor, because it offers ideal conditions for the synthesis of a WSSV subunit vaccine (Medina-Félix et al. 2014).

Synechocystis sp. PCC6803 is one of the most extensively investigated species of photosynthetic single-celled cyanobacterium to be created in a phototrophic cell factory for biomaterials, commodity chemicals, and biofuels. According to studies, the organism *Synechocystis* sp. PCC6803 may

Fig. 1 Graphical abstract illustrates the potential future directions for microalgae-based WSSV management, including integrated approaches with vaccines, transgenic food, and drug supplements for more sustainable aquaculture



efficiently remove phosphate from shrimp effluent and can be fed directly to juvenile shrimp with good security (Zhai et al. 2021).

Synechococcus is a unicellular cyanobacterium with a size range of 0.8–1.5 µm. The unicellular cyanobacterium PCC 7942 (Zhu et al. 2020), also known as *Synechococcus elongatus* PCC 7942, is a model photosynthetic prokaryote extensively studied for its circadian rhythms, photosynthesis, and genetic tractability, making it a valuable organism for both basic biological research and biotechnological applications (Peng et al. 2022). *Anabaena* is a filamentous cyanobacteria found in plankton that can fix nitrogen through specialized cells called heterocysts, which provide a micro-oxic environment necessary for the functioning of nitrogenase, the enzyme responsible for nitrogen fixation (Tong et al. 2023).

This review summarizes the prophylactic activities of microalgae such as *Chlamydomonas reinhardtii*, *Dunaliella salina*, *Synechocystis species*, *Synechococcus elongatus*, *Synechococcus species*, and *Anabaena* species for treating WSSV using various protocols including transgenic algae, recombinant protein, and dsRNA (Table 1).

Chlamydomonas reinhardtii

Antiviral VP-28 dsRNA in the *Chlamydomonas reinhardtii* chloroplast

This study demonstrated an advancement in the generation of microalgal antiviral dsRNA for preventing disease in *penaeid* shrimp. The VP28 gene of the WSSV was the target of 307 bp of sequence in the new vector p2XTRBL, which was then subcloned into the transformation vector pSS116 via Golden Gate assembly (Charoonnart et al. 2022).

A recombinant plasmid was used to transform the *Chlamydomonas reinhardtii* chloroplast, resulting in the production of VP28 dsRNA. PCR analysis confirmed the presence

of the dsRNA-expressing cassette in the algal transformants. Transcript analysis showed that the microalgae produced a significant amount of dsRNA, which was then fed to *Penaeus vannamei* shrimp post-larvae to protect them from WSSV infection. The survival rate of shrimp treated with dsRNA-expressing *C. reinhardtii* after the WSSV challenge was significantly increased (95.2%) compared to the control group. This study demonstrates the potential of using genetically modified microalgae to protect against viral infections in shrimp farming practices (Charoonnart et al. 2022).

Production of an oral vaccine for WSSV using *Chlamydomonas reinhardtii*

In the majority of developing nations, shrimp are an essential part of aquaculture farming, which generates a lot of revenue but also presents additional challenges, such as viral infections, particularly WSSV. (Thorner et al. 2020). Numerous anti-WSSV techniques are still employed to identify the structural protein and develop WSSV vaccines, including recombinant, DNA, and RNA vaccines. *Chlamydomonas reinhardtii* is a single-cell green alga and it is genetically traceable so we can use it as a model organism for molecular studies to understand diseases (Sasso et al. 2018).

In this study, recombinant *Chlamydomonas reinhardtii* expressing the VP28 protein as an oral vaccine against WSSV was successfully developed. The VP28 gene was optimized and cloned into an expression vector, and the recombinant protein was produced in *C. reinhardtii* (Ma et al. 2020). The transgenic algae were fed to *Litopenaeus vannamei* shrimp, and the mortality rate was significantly lower in the transgenic group compared to the control groups. Real-time PCR analysis showed high expression levels of the VP28 gene in the transgenic shrimp. Immunological parameters such as lysosome, anti-lipopolysaccharides, superoxide dismutase, and prophenoloxidase showed increased expression after 14

Table 1 Different types of microalgae treatment methods and their survival rate against white spot syndrome virus infection

Microalgae	Treatment methods	Survival rate (%)	Reference
<i>Chlamydomonas reinhardtii</i>	Antiviral activity using dsRNA in chloroplast	95.2	Charoonnart et al. 2022
	Antiviral activity using transgenic microalgae	70	Thorner et al. 2020
	Oral vaccination	87	Kiataramgul et al. 2020
<i>Dunaliella salina</i>	Vaccination using transgenic microalgae	59	Feng et al. 2014
	Evaluation of productive response to WSSV	56	Medina-Félix et al. 2014
<i>Synechocystis sp</i>	Trans vp28 in WSSV resistance	26.4	Zhai et al. 2021
<i>Synechococcus sp</i>	Immune response against WSSV	71.1	Zhu et al. 2020
	Antiviral activity using transgenic microalgae	71	Peng et al. 2022
<i>Anabaena</i>	Combination of food and drug against WSSV	68	Jia et al. 2016
	Antiviral activity using vp28 protein	45.6 & 62.2	Tong et al. 2023

days, indicating the potential of the recombinant protein to act against WSSV infection (Lanh et al. 2021).

Treatment of WSSV using transgenic microalgae

To control WSSV infectivity, various biological forms of VP28, including DNA vaccines, interfering RNAs targeting vp28 expression, and recombinant protein, were used. But, most of these control measures depend on injecting individually which is difficult in commercial production. Moreover, the cost and time required to prepare VP28 as a recombinant protein, dsRNA, or DNA vaccine are considered additional barriers to the production and application of these agents in aquaculture (Kiataramgul et al. 2020).

Transgenic microalgae were developed to synthesize viral antigens against white spot syndrome virus (WSSV) and were found to be effective in controlling WSSV when delivered orally (Wannathong et al. 2016). The *Chlamydomonas reinhardtii* strains were maintained under certain conditions and the vp28 protein encoding sequence was optimized for compatibility with *C. reinhardtii* (Economou et al. 2014). The transformation of the algae was achieved by introducing the chloroplast expression vectors into *C. reinhardtii* using glass bead transformation (Stern and Harris et al. 2009). The transformed colonies were tested for gene expression and recombinant protein expression. The transgenic algae were found to survive in the digestive system of shrimp and showed a high survival rate against WSSV when administered orally (Kiataramgul et al. 2020).

Dunaliella salina

Developing transgenic *Dunaliella salina* for crayfish white spot syndrome virus vaccination

The WSSV subunit vaccine was produced in the current investigation using a unique expression system called *D. salina*. PPT-resistant *D. salina* transformants were produced following the transformation. Successful insertion of the WSSV VP28 gene into the genome of *D. salina* cells led to the production of a protein with inherent immunological action. The Ds-VP28-vaccinated crayfish had around 59% survival rate, compared to 100% mortality in the Ds-empty and positive control groups. This finding demonstrates that an oral vaccine employing transgenic *D. salina* can protect against WSSV in crayfish (Feng et al. 2014).

Evaluation of *Litopenaeus vannamei*'s productive and physiological responses to WSSV infection and diets enriched with *Dunaliella* sp.

The productive and physiological reactions of *Litopenaeus vannamei* exposed to WSSV and fed diets supplemented

with *Dunaliella* sp., a bacterium with a high β -carotene content generated by nitrogen reduction in the culture medium, were examined experimentally. 1% and 2% microalgae meal in the experimental diets T1 and T2, as well as a baseline diet with 35% crude protein, were assessed. Additionally, positive (infected juveniles) and negative (not infected) controls assessed were (Medina-Félix et al. 2014).

In comparison to the positive control group's survival rate (56%), the two treatment groups' survival rate (80%) was much greater. Survival in the adverse control group was 100%. Although the tendencies were not clear, some metabolites (glucose and triglycerides) appeared to decrease on the final days of the trial, likely because of their use as energy for the shrimp to thrive despite the infection. Some variation was observed for hemolymph metabolites between treatments and at different times post-infection. The study's findings point to a beneficial effect of including *Dunaliella* meal in the diet on shrimp survival and an ambiguous impact on hemolymph metabolites (Medina-Félix et al. 2014).

The significance of the trans-VP28 gene in white spot syndrome virus (WSSV) resistance in *Synechocystis* sp. PCC6803

The study investigated the impact of the trans-VP28 gene from *Synechocystis* sp. PCC6803 on *Penaeus vannamei* in combating white spot syndrome virus (WSSV) by analyzing immune-related enzyme activities and histopathological changes. Shrimp were fed *Synechocystis* sp. containing the trans-VP28 gene for 10 days before being exposed to WSSV. Enzyme activity, survival rates, and tissue histology were assessed. Results showed that 7 days after the WSSV challenge, the experimental group (fed with trans-VP28 *Synechocystis* sp. cells) had a survival rate of 26.4%, while the wild-type group (fed with wild-type *Synechocystis* sp. cells) had a survival rate of 80.2%. The positive control group (challenged but not vaccinated) experienced 100% mortality, while the negative control group (neither challenged nor vaccinated) had no fatalities. Following vaccination, the experimental group exhibited peak activities of alkaline phosphatase (AKP), peroxidase (POD), and acid phosphatase (ACP) in both skeletal muscle and hepatopancreas tissues. The enzyme activities in the experimental group were higher than those in the wild-type and positive control groups, indicating that the trans-VP28 gene enhances immune response and provides some protection against WSSV (Zhai et al. 2021).

Histopathology analysis of shrimp tissues revealed distinct differences among experimental groups. In the positive control group, muscle fibers appeared wavy, with enlarged and irregularly distributed nuclei. Conversely, the wild-type group also had wavy muscle fibers, but with less pronounced

nuclear abnormalities. The negative control group showed neatly structured muscle fibers with evenly distributed nuclei. In the hepatopancreas samples, the negative control exhibited small cell gaps and well-organized nuclei. However, the positive control had severe damage, including degenerated cells and damaged cell walls and connective tissue. The experimental groups, including those with the trans-VP28 gene from *Synechocystis* sp., had slightly looser connective tissue but well-organized cells and nuclei. These results suggest that the trans-VP28 gene might enhance shrimp immunity (Zhai et al. 2021).

Synechococcus

WSSV infection in *Synechococcus* sp. PCC 7942 with VP 28, VP19, and VP (19 + 28) genes: the immunological response and impacts of *L. Vannamei*

Synechococcus is a unicellular cyanobacterium with a size range of 0.8–1.5 μm . In this study, researchers successfully cloned VP19 and VP (19 + 28) into a cyanobacterium called *Synechococcus* sp. PCC 7942 to create mutant strains. The gene expression levels of these mutants were found to be higher compared to the wild type. When these mutants were used in vaccination trials with shrimp infected with WSSV, the survival rates were significantly higher in the mutant groups compared to the positive control group. Enzymatic activity analysis showed that the mutants had different effects on the activity levels of various enzymes compared to the positive control group. This study demonstrates the potential antiviral activity of the mutant *Synechococcus* strains in protecting shrimp against WSSV infection (Zhu et al. 2020).

VP 18 and mOrange gene used in the construction of transgenic *Synechococcus elongatus* PCC 7942 against WSSV

In a study on *Synechococcus elongatus* microalgae, researchers aimed to explore its potential as a vector for expressing VP28 and mOrange genes to combat white spot syndrome virus (WSSV). The genes were cloned into an expression vector, pRL-489, and transferred to *E. coli* Top10. Confirmation was done through PCR, restriction digestion, and Western blotting. Gene expression was assessed by RT-qPCR over 15 days, showing peak expression on day 6. Immunization experiments were conducted on *L. vannamei* shrimp, with eight experimental groups, including positive and negative controls, wild-type, VP28, and various VP28–mOrange fusion constructs. The positive control group experienced 100% mortality within 6 days, while the

negative control group had 100% survival. Survival rates in the VP28–mOrange and VP28–mOrange fusion groups ranged from 33.3 to 81%. Enzymatic activity assays for SOD, CAT, and FAS indicated increased activity in the wild-type and positive control groups within 12 h. The positive control showed the highest levels of PO activity compared to other groups (Peng et al. 2022).

Anabaena

Combination of drug and food supplements using *Anabaena* sp. PCC 7210 against WSSV

Anabaena is a filamentous cyanobacterium commonly present in plankton and nitrogen-rich habitats, where it plays a key role in nitrogen fixation by forming heterocysts—specialized cells that enable the conversion of atmospheric nitrogen into a usable form. The cloning of VP28 is done in the vector pUC-18 and then transferred into *Anabaena* PCC 7120. There are two mutants: transgenic *Anabaena* mutant PRL-489-VP28 and blank by PRL-489. The mutant and wild-type RNA were extracted and converted into cDNA to analyze the gene expression by RT-qPCR, followed by western blotting. The vaccination experiment was performed on post-larvae *L. vannamei* in five groups: positive control, negative control, wild type, blank, and mutant VP28. The net photosynthesis was calculated using a Clark-type oxygen electrode (Jia et al. 2016).

VP28 and WSSV DNA were successfully cloned, the vector containing VP28 was confirmed using restricted endonuclease, and PCC 7210 was transferred into blank and transgenic mutant cells. *Anabaena* mutant caused by VP28 showed a significant increase in gene expression, with the copy number reaching 1.99×10^5 copies mL^{-1} on day 15, and the VP28 protein concentration reaching 34.5 mg L^{-1} , indicating successful expression and accumulation of the target protein over time. The vaccination challenge lasted 10 days and resulted in 100% mortality in the positive control, 15.5% survival in the wild type, 20.6% survival in the blank treatment, and 68% survival in the mutant transgenic. The photosynthesis activity was 24.7–42.7 times higher than in the wild type (Jia et al. 2016).

Anabaena sp. PCC 7210 expression with VP28 protein against WSSV analysis of survival and histopathology

The experiment involved feeding different groups of shrimps with different preparations of Ana 7210, including wild type, empty vector, and Ana 7210 pRL-489-VP28. The shrimp were divided into six groups with 30 shrimp in each group, and each group was fed a specific percentage of the Ana

7210 preparations. Samples were collected during the experiment and tested using RT-qPCR to determine the activity of WSSV in various shrimp tissues, such as gills, muscles, and hepatopancreas. Additionally, histopathology was performed on the shrimp that survived until the 4th day of the experiment, focusing on organs such as hepatopancreas, gills, and tail muscles. Overall, the study aimed to investigate the effects of feeding shrimp with different preparations of Ana 7210 on WSSV activity and the histopathological changes in shrimp tissues (Tong et al. 2023).

The viral challenge resulted in 100% mortality for the positive control, 81.1% for the wild group, 86.7% for the empty vector group, and 45.6% and 62.2% for the food supplement group, respectively. In immunity 1 and immunity 2, the empty vector copy number was lowered to 106 copies/L and 105 copies/L, respectively, whereas the positive control copy number remained at 108 copies/L in the wild type. The hepatopancreas histopathology investigation revealed smudged cells in immunity group 1 and normal cells in immunity group 2, along with no change in the negative control and damage in the positive control. The positive control gill filament was damaged, appearing bloated and rupturing in immune group 1, but the negative control gills remained unchanged. Normal, whole shrimp organs were also visible (Tong et al. 2023).

Discussion

The development and effectiveness of *C. reinhardtii* dsRNA expressing lines have significantly improved the novel vector p2xTRBL and its related transformation vector pSS116. It should be noted that the p2xTRBL cassette is also the first completely designed method for *C. reinhardtii* dsRNA expression. By modifying pSRSapI to include convergent *psaA* promoters, Charoonnart et al. (2022) previously created a dsRNA-expressing cassette. However, the original vector was made for protein synthesis, it also contained extraneous 5'UTR regions, and only the sense promoter had a corresponding terminator. In contrast, the antisense promoter continues to drive transcription until it reaches a suitable native terminator, giving the dsRNA a lengthy ssRNA "tail" and wasting cellular resources. Being prokaryotic in origin, chloroplast promoters can also be expressed in *E. coli*, and the *rrnS* promoter is no exception. Before the algal transformation, this provided a helpful opportunity to confirm the pSS116 VP28 dsRNA-expressing cassette and to create a stock of VP28 dsRNA that could be utilized as a positive control. In aquaculture, conventional antibiotic resistance selection markers are strictly restricted to prevent the horizontal gene transfer of such cassettes. Therefore, it is highly advantageous that chloroplast transformants can be chosen without the use of such markers, as has been shown

repeatedly. Although the amount of dsRNA obtained was far smaller than that of the bacterial system, it was still worthwhile because the transgenic cell could be utilized as direct feed and the system was markerless. According to data gathered from farms in Southern Thailand by the Department of Fisheries, the *Penaeus vannamei* shrimp is known to be extremely susceptible to WSSV throughout its life cycle, but the post-larval (PL) stage of growth is the most vulnerable to viral infection, often resulting in high mortality rates and significant economic losses for shrimp farmers due to the rapid spread and severity of the disease during this critical developmental phase. Therefore, it is crucial to confirm that PL stocks are in good health and free of disease before moving them to the grow-out pond. As WSSV can spread horizontally, WSSV transmission in the TN72-SS group may have been quicker than in the TN72-VP28 group, leading to a faster rate of 100% mortality. The huge mortality of shrimp due to viral infection was deterred by feeding dsRNA-producing microalgae.

Oral vaccination was performed using the VP28 gene, which was cloned into the pChlamy4 expression vector, and the construct was transformed into *E. coli* DH5 α cells. The recombinant plasmid pChlamy4_VP28 was successfully constructed, followed by the expression of the VP28 recombinant protein in *C. reinhardtii*. *C. reinhardtii* was confirmed through molecular analyses, including RT-PCR, Western blotting, and ELISA. Cloning of the vector was also successfully verified. Shrimp fed these transgenic algae exhibit an 87% survival rate against the white spot syndrome virus. Algal cells are capable of surviving in the digestive system of shrimp, which makes these a better oral delivery system, and along with some of the VP28 released sufficient for stimulating the shrimp's immunity. It is suggested as a sustainable vaccine production method that is environmentally friendly. This approach is proposed as a sustainable and environmentally friendly method for vaccine production. Based on the supporting evidence, it shows potential not only for preventing WSSV infections but also for combating other viral diseases in aquaculture.

Through oral administration, transgenic *D. salina* strains successfully produced the WSSV VP28 gene and generated protection against WSSV in crayfish. According to the study by Feng et al., it is possible to produce an oral vaccine with *D. salina* that can be used to prevent WSSV infections as well as manage other viral diseases in crustacean species. In the second study, the survival of juvenile *L. vannamei* was improved by the addition of *Dunaliella* meal to the diet at the two levels examined in productive and physiological responses to WSSV infection and diets. Metabolites and certain effects on hemolymph were reported. In crustaceans, the hepatopancreas is a vital immunological organ. According to reports, the innate immunity in *Penaeus vannamei* and white

shrimp *Litopenaeus setiferous* is significantly influenced by the hepatopancreas. Several antioxidant enzymes, including superoxide dismutase, catalase, and phenoloxidase, can have their activity improved by the trans-VP28 gene found in *Synechocystis* sp. PCC6803, but it can also help *Penaeus vannamei* flourish.

Microalgae are important in the fight against WSSV. Using the genes VP19 and VP (19 + 28) in *L. vannamei*, *Synechococcus* sp. PCC 7942 can act against WSSV. The VP28 and mOrange genes' protective effects are not reduced, and they are still greater than that of transgenic VP28. *Anabaena* sp. PCC 7120 expressed with VP28 does not have significant antiviral activity, but it does improve growth, and *Anabaena* sp. PCC 7120 expressed with VP28 has a significant survival rate in the immunity group. Unlike antibiotics, microalgae are environment friendly and can be employed for controlling other aquatic viral diseases based on demand.

Conclusion

The development of dsRNA-expressing lines in *C. reinhardtii* has improved the novel vector p2xTRBL and its transformation vector pSS116, providing a markerless method for selecting transgenic cells. This has been particularly effective in producing VP28 dsRNA for combating white spot syndrome virus (WSSV) in shrimp. Recombinant plasmids have been created to express VP28 genes in microalgae, leading to successful oral vaccines that have shown high survival rates in shrimp against WSSV. Additionally, transgenic *Dunaliella salina* strains produced VP28 genes to protect crayfish from WSSV. Microalgae, such as *Synechococcus* sp. and *Anabaena* sp., were utilized for their antiviral properties against WSSV, showing promise in controlling other aquatic viral diseases. Overall, using microalgae as oral vaccines presents an environmentally friendly and sustainable method for disease prevention in aquaculture.

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Data availability All authors declare that all data and materials as well as the software applications support their published claims and comply with the field standards.

Declarations

Conflict of interest There is no conflict of interest.

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